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Horizon Entropy and the Area Law

How a formula first written down for black holes also fixes an entropy scale for the universe

By Lee McCulloch-James

When a region of spacetime is bounded by a horizon, physicists can assign that horizon a definite entropy. The entropy depends on its surface area measured in units of the Planck length squared. First developed for black holes in the early 1970s, the relation also applies to cosmological horizons. It provides a remarkably direct route from the geometry of the universe to an entropy scale of order $10^{122}k_B$.

From Black Hole Physics

In 1972, John Wheeler asked his doctoral student Jacob Bekenstein what happens to the entropy of a cup of tea after it falls into a black hole. Ordinary thermodynamics forbids entropy from simply vanishing, yet a classical black hole was thought to be featureless, retaining no record of what fell in. Bekenstein proposed that the black hole itself carries entropy proportional to the area of its event horizon (Bekenstein, 1973).

Hawking initially objected. A body with entropy should have a temperature and radiate, while a classical black hole was believed to absorb everything and emit nothing. His 1975 calculation using quantum field theory in curved spacetime showed that black holes do radiate, with a temperature set by the horizon's surface gravity (Hawking, 1975). That calculation fixed the exact coefficient in Bekenstein's formula. Bardeen, Carter, and Hawking had already shown in 1973 that black holes obey laws with the same mathematical form as the four laws of thermodynamics (Bardeen, Carter, and Hawking, 1973). Hawking's radiation calculation gave those thermodynamic quantities a physical temperature and entropy.

Two Ingredients, Cleanly Separated

The Bekenstein–Hawking law can be written as

$$S_H = \sigma_H A,$$

where

$$\boxed{\sigma_H \equiv \frac{k_B c^3}{4G\hbar} = \frac{k_B}{4\ell_P^2}}$$

and

$$\ell_P = \sqrt{\frac{G\hbar}{c^3}}$$

is the Planck length. The constant σ_H is fixed by $k_B, \hbar$, c, and G. It specifies the entropy associated with each unit of horizon area. The area A supplies the geometry of the particular horizon. It is standard in this subject to work in natural units, setting $c = G = \hbar = 1$. In these units $\ell_P = 1$, and the law becomes

$$\boxed{\frac{S_H}{k_B} = \frac{1}{4}A}$$

with area measured in Planck areas. Every physical constant has been absorbed into the choice of units. The remaining statement is geometric: the entropy in units of k_B is one quarter of the horizon area in Planck units.

BY THE NUMBERS

Planck length ℓ_P	1.616×10^{-35} m
Boltzmann constant k_B	1.381×10^{-23} J/K
Hubble parameter H_0	67.4 km/s/Mpc
Hubble radius R_H	14.5 Gly
Cosmic horizon entropy	$2.6 \times 10^{122}\, k_B$
Entropy inside horizon	$10^{103\text{–}104}\, k_B$

Cosmological Horizons

An observer in an accelerating universe such as ours is also associated with a cosmological horizon. Gibbons and Hawking showed in 1977 that a de Sitter cosmological horizon obeys the same entropy–area law (Gibbons and Hawking, 1977).

For pure de Sitter spacetime, an idealized universe whose expansion is driven by a cosmological constant Λ, the horizon radius is

$$R_{\text{dS}} = \frac{c}{H}.$$

The Friedmann equation gives

$$H^2 = \frac{\Lambda c^2}{3},$$

so

$$R_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad A_{\text{dS}} = \frac{12\pi}{\Lambda}.$$

Applying the area law while keeping the universal constant and the geometry separate gives

$$\boxed{S_{\text{dS}} = \underbrace{\sigma_H}_{k_B c^3/4G\hbar} \underbrace{\frac{12\pi}{\Lambda}}_{A_{\text{dS}}} = \frac{3\pi c^3}{G\hbar}\frac{k_B}{\Lambda}}$$

The dependence on Λ therefore enters through the geometry:

$$\boxed{S_{\text{dS}} \propto A_{\text{dS}} \propto R_{\text{dS}}^2 \propto H^{-2} \propto \Lambda^{-1}.}$$

A small cosmological constant corresponds to a large horizon, a large horizon area, and consequently a large horizon entropy.

A Universe-Sized Number

The present Hubble parameter inferred from the Planck satellite's measurement of the cosmic microwave background is

$$H_0 \approx 67.4\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$$

(Planck Collaboration, 2018), giving a Hubble radius of about 14.5 billion light-years. The real universe follows the ΛCDM model rather than pure de Sitter spacetime, so although the present Hubble radius and cosmological event horizon are distinct the Hubble radius provides an appropriate order of magnitude for this calculation.

With

$$R_H \sim 10^{26}\ \text{m}, \qquad \ell_P \approx 1.616 \times 10^{-35}\ \text{m},$$

the area law gives

$$\boxed{\frac{S_H}{k_B} \sim \left(\frac{R_H}{\ell_P}\right)^2 \sim 10^{122}.}$$

A more detailed treatment gives a cosmological horizon entropy of approximately $2.6 \times 10^{122}\, k_B$ (Egan and Lineweaver, 2010).

This enormous number is already contained in the geometry. In Planck units the observed cosmological constant is of order 10^{-122}, while de Sitter entropy scales as its inverse. The appearance of an entropy scale near $10^{122} k_B$ therefore follows directly from the measured cosmological scale.

The Tempting Shortcut

There is a temptation to instead of treating the horizon geometrically, take all the mass contained within a Hubble-sized sphere and insert it into the black-hole entropy formula. Thus at the critical density

$$\rho_c = \frac{3H^2}{8\pi G},$$

the mass inside a sphere of radius

$$R_H = \frac{c}{H}$$

is

$$M_H = \rho_c \frac{4}{3}\pi R_H^3 = \frac{c^3}{2GH}.$$

Now calculate the Schwarzschild radius associated with that mass:

$$R_s = \frac{2GM_H}{c^2} = \frac{c}{H} = R_H.$$

Thus

$$\boxed{R_s = R_H.}$$

Feeding M_H into the Bekenstein–Hawking mass formula consequently produces the same area and the same entropy:

$$\boxed{S_{BH}(M_H) = S_H \sim 10^{122} k_B.}$$

The calculation returns to the horizon from which it began: at critical density, the mass contained within the Hubble sphere carries precisely the mass scale whose Schwarzschild radius equals the Hubble radius. This equality is built into the definition of critical density; it does not mean that the Hubble sphere is literally a Schwarzschild black hole. Treating that entire mass with the black-hole formula therefore reproduces the entropy associated with the boundary.

This gives us two different quantities to keep track of:

$$\boxed{S_{\text{horizon}} \sim 10^{122} k_B}$$

and

$$\boxed{S_{\text{contents}} = \sum_i S_i.}$$

The first follows from the geometry of the horizon whilst the second has to be built from the physical contents of the universe.

Anticipating the count assembled later: the best present inventory of stars, stellar remnants, and black holes comes to roughly $10^{104} k_B$ (Egan and Lineweaver, 2010), about eighteen orders of magnitude below S_{horizon}. The smallness of Λ explains why the horizon entropy scale itself is so large. The size of the interior inventory is a separate question, set by the structures that cosmic history has actually produced.

The Cosmological Constant Problem

In Planck units, Λ is measured to be of order 10^{-122}. Quantum field theory's naive estimate of the vacuum energy differs from the observed value by around 120 orders of magnitude, a discrepancy discussed in Weinberg's influential 1989 review and still unresolved (Weinberg, 1989).

The area law does not determine why Λ has its observed value. Once that value is given, however, the relation

$$S_{\text{dS}} \propto \frac{1}{\Lambda}$$

fixes the corresponding de Sitter entropy scale. The question can now change. Having obtained the geometric scale S_{horizon}, how much entropy is actually carried by the contents of the universe?

A Family Resemblance

The Stefan–Boltzmann law provides a familiar starting point for following the ordinary contents of the universe. For a radiating surface of area A at temperature T, the total power emitted is

$$P = \sigma_{\text{SB}} A T^4, \qquad \sigma_{\text{SB}} = \frac{\pi^2 k_B^4}{60\hbar^3 c^2},$$

and the intensity is

$$I \equiv \frac{P}{A} = \sigma_{\text{SB}} T^4.$$

There is a useful structural resemblance to $S_H = \sigma_H A$.

$$\sigma_{\text{SB}} = \frac{\pi^2 G k_B^3}{15\hbar^2 c^5} \sigma_H$$

Both laws combine a universal constant with an area. The horizon constant σ_H also contains G, bringing gravitation and spacetime geometry into the thermodynamic description.

The cosmic microwave background as a real blackbody with temperature $T = 2.725$ K (Fixsen, 2009) described by the Stefan–Boltzmann law , has intensity

$$I = \sigma_{\text{SB}} T^4 \approx 3.1 \times 10^{-6}\ \text{W m}^{-2}.$$

Wien's displacement law,

$$\lambda_{\text{max}} T = b, \qquad b = 2.898 \times 10^{-3}\ \text{m K},$$

gives

$$\lambda_{\text{max}} \approx 1.06\ \text{mm}.$$

Converting through

$$E = \frac{hc}{\lambda_{\text{max}}}$$

gives a characteristic photon energy approximately 1.2×10^{-3} eV.

Star by Star

Whilst the Stefan–Boltzmann law is one of the power laws governing a star's energy output, Luminosity also follows an empirical mass–luminosity relation, roughly

$$L \propto M^{3.5}$$

for Sun-like stars, flattening toward $L \propto M$ for the most massive stars. A star's available nuclear fuel is a fraction of Mc^2, so its main-sequence lifetime scales approximately as

$$t \sim \frac{Mc^2}{L}.$$

The lifetime therefore falls steeply with increasing mass, from roughly ten billion years for the Sun to a few million years for the most massive stars.

The Sun provides a useful order of magnitude input with luminosity,

$$L_\odot = 4\pi R_\odot^2 \sigma_{\mathrm{SB}} T_\odot^4 \approx 3.83 \times 10^{26}\ \mathrm{W}.$$

Multiplying by its roughly ten-billion-year main-sequence lifetime gives

$$Q_\odot = L_\odot t_\odot \approx 1.2 \times 10^{44}\ \mathrm{J}.$$

Relative to its rest energy we have,

$$\frac{Q_\odot}{M_\odot c^2} \approx 6.8 \times 10^{-4},$$

which is consistent with the 0.7% mass-to-energy efficiency of hydrogen fusion combined with the fact that only a fraction of the Sun's mass participates in core hydrogen burning.

Scaled very roughly to the universe's 10^{22} to 10^{24} stars, all starlight ever radiated corresponds to an energy of order

$$10^{67}\ \mathrm{J}.$$

Taking a representative emission temperature of $T \sim 6000$ K, close to the Sun's photosphere, the relation $\Delta S = Q/T$ gives a crude source-entropy scale

$$S_{\text{stellar radiation}} \sim 10^{86} k_B.$$

This is not a present-day entropy census of the radiation field, which has subsequently expanded and cooled. It is only an order-of-magnitude estimate for the entropy associated with stellar emission at its source. Against the horizon scale of $10^{122} k_B$, this stellar-radiation scale is already very small.

A NOTE ON UNITS

The ideal gas law states

$$pV = N k_B T,$$

with N the dimensionless number of particles. Pressure times volume has units of energy. Since N is a pure number and T is measured in kelvin, k_B carries units of $\mathrm{J\,K^{-1}}$. Those are the units of entropy. The thermodynamic definition $\Delta S = Q/T$ gives the same units. Boltzmann's statistical definition,

$$S = k_B \ln \Omega,$$

makes the role of k_B particularly clear: Ω is a pure count, so k_B supplies the units. The horizon law has the same dimensional structure:

$$S_H = k_B \frac{A}{4\ell_P^2}.$$

The ratio $A/4\ell_P^2$ is dimensionless, and k_B supplies units of entropy.

Black Holes Change the Accounting

The law that opened this piece was first derived for black holes, then carried outward to the cosmological horizon. It can now be brought back to where it started: the black holes actually contained within the universe. Written in terms of mass, the Bekenstein–Hawking formula is

$$\frac{S_{BH}}{k_B} = \frac{4\pi G M^2}{\hbar c}.$$

Setting M equal to one solar mass fixes constant of proportionality:

$$\boxed{\frac{S_{BH}}{k_B} \approx 1.05 \times 10^{77} \left(\frac{M}{M_\odot}\right)^2.}$$

A single Sun-mass black hole, on its own, already carries $1.05 \times 10^{77} k_B$ of entropy. Set that against the star that made it. Over ten billion years the Sun will radiate roughly $10^{63} k_B$ of entropy into space, and the matter it is built from carries less still: about 1.2×10^{57} baryons at one to a few k_B of entropy per baryon, some 10^{57} to $10^{58} k_B$ in total. One horizon holds some fourteen orders of magnitude more entropy than its progenitor's entire radiative output, and twenty more than the matter itself.

That gap can look perverse, and the resolution is not that stars are inefficient. Writing $\varepsilon \approx 6.8 \times 10^{-4}$ for the lifetime fusion efficiency found above, and T_H for the Hawking temperature, the ratio factors cleanly:

$$\frac{S_{BH}}{S_{\text{rad}}} = \frac{3}{8}\,\underbrace{\frac{1}{\varepsilon}}_{\sim 10^3}\,\underbrace{\frac{T_\star}{T_H}}_{\sim 10^{11}}.$$

Restoring the fuel a star never burns closes barely three of the fourteen orders. The rest is temperature: entropy is Q/T, and a star dumps its energy at a photospheric ~ 5800 K whereas a solar-mass horizon radiates at $T_H \approx 6 \times 10^{-8}$ K. Even a perfectly efficient star, converting its whole rest mass to light at 6000 K, would fall eleven orders short. Collapse wins not because stars waste their fuel but because a horizon is an extraordinarily cold place to put energy.

Collapse does not destroy the star's entropy; the second law demands the horizon carry at least as much, and the area law shows it carries far more. That requirement is what forced Bekenstein's hand, and what Hawking's temperature made rigorous rather than mere bookkeeping. Whether the star's microscopic detail survives in the radiation that eventually carries the black hole away, or is genuinely lost, is the open question revisited at the end of this piece.

The important feature is the square:

$$\boxed{S_{BH} \propto M^2.}$$

Large black holes therefore dominate the entropy inventory very quickly.

The progenitors of stellar-mass black holes are short-lived massive stars. A 2022 study estimated that there are of order 4×10^{19} stellar-mass black holes in the observable universe (Sicilia et al., 2022). Taking $30 M_\odot$ as a representative mass gives

$$S_{\text{stellar BH}} \sim 4 \times 10^{19} \times 1.05 \times 10^{77} \times 30^2\, k_B \sim 4 \times 10^{99}\, k_B.$$

Supermassive black holes are much less numerous. Taking roughly one per major galaxy, of order 10^{11} objects, and a representative mass $10^8 M_\odot$ gives

$$S_{\text{SMBH}} \sim 10^{11} \times 1.05 \times 10^{77} \times (10^8)^2\, k_B \sim 10^{104}\, k_B.$$

The M^2 dependence makes the result decisive. Stellar-mass black holes may outnumber supermassive black holes by hundreds of millions to one, yet the supermassive population dominates the entropy. This simple estimate lands close to Egan and Lineweaver's 10^{103} to $10^{104}\,k_B$ estimate for the entropy of the contents inside the cosmological horizon.

How Full Is the Entropy Budget?

The detailed count assembled above confirms what was anticipated earlier:

$$\boxed{S_{\text{horizon}} \sim 10^{122} k_B, \quad S_{\text{contents}} \sim 10^{104} k_B, \quad \frac{S_{\text{horizon}}}{S_{\text{contents}}} \sim 10^{18}.}$$

Geometry fixes the first number, through the horizon's dependence on Λ. Cosmic history, the formation of stars, stellar remnants, and especially supermassive black holes, fixes the second.

From the Entropy Bound to Dark Energy

Throughout this piece Λ has appeared as a geometric parameter setting the size of the horizon. Physically it is usually identified with dark energy, a smooth component that today makes up about 68.5% of the universe's critical density, $\Omega_\Lambda \approx 0.685$ (Planck Collaboration, 2018), and drives the acceleration of the expansion.

The difference between S_{horizon} and S_{contents} should not be read as a reservoir of missing entropy waiting to be filled. Horizon entropy, matter entropy, black-hole entropy, and dark-energy density are distinct quantities. A related line of research instead asks how the same horizon-based bound might constrain the amount of vacuum energy contained within a region of cosmic size.

Demanding that the vacuum energy inside a sphere of radius R_h not exceed the mass-energy of a black hole of that size gives, in one class of models,

$$\boxed{\rho_\Lambda \sim n^2 \rho_c}$$

where $\rho_c = 3H^2/8\pi G$ is the same critical density used earlier and n is a free parameter expected to be of order one (Li, 2004). Setting $n \sim 1$ predicts a dark energy density comparable to the critical density itself, consistent at the order-of-magnitude level with the observed $\Omega_\Lambda \approx 0.685$, and without invoking the 120-order-of-magnitude mismatch that afflicts the naive quantum field theory estimate.

This is one proposed answer to why Λ takes the value it does, not a settled one; it remains actively debated. It shows, though, that the same reasoning that opened this piece, bounding a region's physical content by its horizon rather than its volume, extends directly into live research on the nature of dark energy itself.

Holography and What Remains Open

The area law has motivated a much broader idea about gravity and information. Gerard 't Hooft and Leonard Susskind proposed in the 1990s that the information associated with a region of space may admit a description on its boundary, an idea now called the holographic principle ('t Hooft, 1993; Susskind, 1995). Juan Maldacena's 1997 work subsequently provided a precise realization of holographic duality in a particular class of spacetimes (Maldacena, 1998).

The cosmological calculation makes the attraction of the idea visible. Count the Planck areas on a horizon, divide by four, and the result is its entropy in units of Boltzmann's constant:

$$\boxed{\frac{S_H}{k_B} = \frac{A}{4\ell_P^2}.}$$

The same rule applies to a black-hole horizon and, in de Sitter spacetime, to a cosmological horizon.

For our universe the resulting geometric entropy scale is of order $10^{122} k_B$. An inventory of the structures currently inside gives roughly $10^{104} k_B$, overwhelmingly concentrated in supermassive black holes. The difference between those numbers is therefore meaningful. The first is set by the geometry of the horizon. The second records the entropy carried by the contents that cosmic history has produced.

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KEY TERMS

Horizon. A causal boundary associated with the limits from which signals can reach an observer or escape a region.
Planck length. The length scale $\ell_P = \sqrt{G\hbar/c^3}$ constructed from G, $\hbar$, and c.
De Sitter spacetime. An idealized spacetime whose expansion is driven by a positive cosmological constant.
Hawking temperature. The temperature of the radiation emitted by a horizon, $T_H \propto 1/M$ for a black hole; 6.2×10^{-8} K for one solar mass.
Holographic principle. The proposal that the information associated with a gravitational region can be described in terms of degrees of freedom associated with a lower-dimensional boundary.
Mass–luminosity relation. The empirical relation between a star's mass and luminosity, approximately $L \propto M^{3.5}$ over part of the main sequence.